



Fillet wire

Round the corners of a wire or edge chain: an exact tangent arc Onshape recognises as an arc, or a smooth curvature blend -- picked corner by corner.



What it does

Onshape can fillet corners in a sketch, but not the corners of a wire. Fillet wire rounds chosen corners of any wire or edge chain -- in 2D or 3D -- and edits a wire body in place, so the untouched parts keep their identity.

Use it when

- A wire (not a sketch) needs rounded corners
- The fillets must be true arcs with a radius
- A 3D corner needs a smooth (G2) blend

W1: two 90 deg corners, r10 (Tangent)

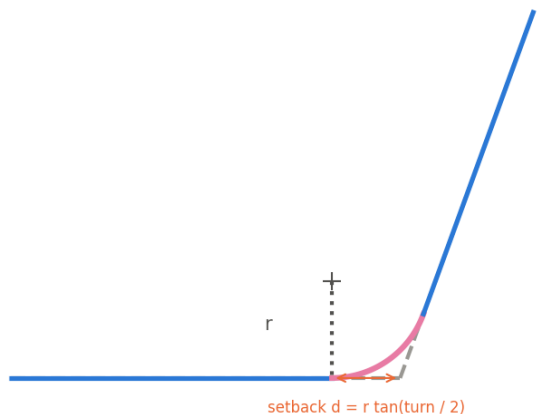


W1: two 90 deg corners filleted r10 (magenta).

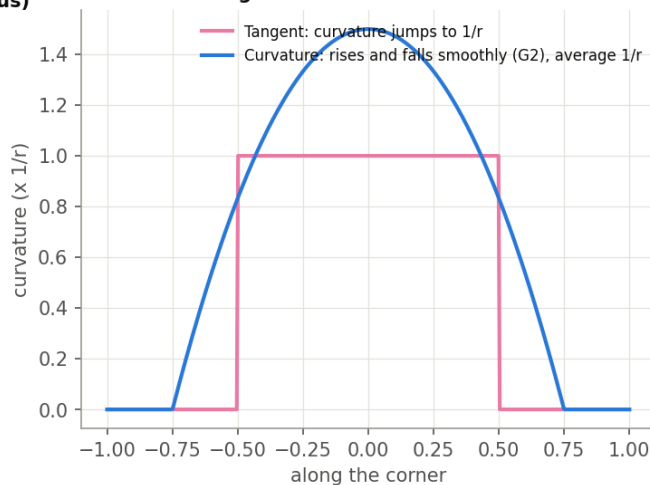


Tangent arc or curvature blend

Tangent: an exact arc (a sketch arc, so Onshape shows its radius)



Tangent arc vs curvature blend



Tangent

An exact arc, set back $d = r \tan(\text{turn} / 2)$, emitted as a sketch arc. Needs the corner to be planar.

Curvature

A G2 blend matching both edges' curvature, sized so its average curvature is $1/r$. Works in 3D.

In place

A wire is split at the tangent points and only the corner pieces are replaced.



The dialog

W2 Wire in place, picked ... ✓ ✗

1 Curves
Test_Wire ✕

2 **Tangent** Curvature

3 Radius 10 mm

4 ☐ Apply to all corners

5 Corners ↑↓ CLEAR
> 5 mm ✕
> 10 mm ✕
> 5 mm ✕
Add Corner

6 ☒ Join into one wire

7 > Advanced
Final

#	Parameter	What it means
1	Curves	Edges or wire bodies.
2	Continuity	TANGENT: exact arcs. CURVATURE: G2 blends.
3	Radius	The fillet radius.
4	Apply to all corners	Off: only the corners you pick.
5	Corners	Filled by clicking corner points in the view; each can have its own radius.
6	Join into one wire	Wire input: rejoin the pieces into one body.
7	Advanced	Corner angle (0.5 deg), Print corners.



Examples

W3: line into an arc, r10 -- tangent to both



W3: a line running into an arc, r10 -- the fillet is an exact circle tangent to both

W1: two 90 deg corners, r10 -> one wire 291.416 mm long ($300 - 4r + \pi r$). W4: a 3D corner with Tangent is skipped (not planar); W5: the same corner with Curvature -> a G2 blend, average curvature 0.1 /mm.



Outputs and messages

Published for Extract variables

Key	What it holds
filletEdges	The fillet edges.
cornerCount / filletCount / skippedCount	Corners found / filleted / not filleted.

Messages

	Message / what to do
Warning	N corner(s) not filleted: ... With the reason: radius too large, no circle touches both edges, not in one plane (use Curvature), edges too short, or fillets overlapping.
Info	N corner(s) found. Click them... Nothing picked yet.



Tips and limits



Radius shown in Onshape

Tangent fillets are sketch arcs, so clicking one shows its radius.



Curvature fillets are splines

They have no single radius.



No end keys yet

Fillet wire does not publish start / end keys.



Where it lives

Document	Curve_tools (2143812a99089658c704f0bc)
Tab	fillet_wire (uses curve_core)
Tests	Fillet wire tests studio: W1-W6 (devtools/onshape/check_fillet_wire.py)
Examples	Fillet wire tests studio
Explainer	docs/explainers/curve_tools/curve_tools_explained.md, section 2.5
Related	Clean wire, Map curve, Merge curve, Evaluate profiles, Fillet wire; Extract variables